

无理由的拖更。

题目 1

求极限：

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{\left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^{n-1}}{n-1}\right)^{n+1}}{\left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}\right)^n} dx$$

解答 1

令 $f_n(x) = 1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}$ ，应用伯努利不等式，当 $0 \leq x \leq 1$ 时，有：

$$\left(\frac{1 + x + \frac{x^2}{2} + \cdots + \frac{x^{n-1}}{n-1}}{1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}}\right)^n = \left(1 - \frac{1}{f_n(x)} \cdot \frac{x^n}{n}\right)^n \geq 1 - \frac{x^n}{f_n(x)}$$

因此

$$\begin{aligned} I_n &\geq \int_0^1 \left(f_n(x) - \frac{x^n}{n}\right) \left(1 - \frac{x^n}{f_n(x)}\right) dx \\ &\geq \int_0^1 \left(f_n(x) - \frac{x^n}{n}\right) dx - \int_0^1 x^n dx \end{aligned}$$

又有

$$\int_0^1 f_n(x) dx = \int_0^1 \left(1 + x + \frac{x^2}{2} + \cdots + \frac{x^n}{n}\right) dx = 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n+1)}$$

同时

$$\int_0^1 \frac{x^n}{n} dx = \frac{1}{n(n+1)}, \quad \int_0^1 x^n dx = \frac{1}{n+1}$$

因此

$$I_n \geq 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} - \frac{1}{n+1} = 2 - \frac{1}{n} - \frac{1}{n+1}$$

另一方面，有

$$I_n \leq \int_0^1 \left(f_n(x) - \frac{x^n}{n}\right) dx = 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} = 2 - \frac{1}{n}$$

得到不等式

$$2 - \frac{1}{n} - \frac{1}{n+1} \leq I_n \leq 2 - \frac{1}{n}$$

当 $n \rightarrow \infty$ 时，左右两边都趋于 2，由夹逼准则得：

$$\lim_{n \rightarrow \infty} I_n = 2$$

题目 2

求

$$\int_0^{\pi} f(x) \mathrm{d} x$$

其中

$$f(x) = \int_0^x \frac{\sin t}{\pi - t} \mathrm{d} t$$

解答 2

$$\begin{aligned} \int_0^{\pi} f(x) \mathrm{d} x &= \int_0^{\pi} \int_0^x \frac{\sin t}{\pi - t} \mathrm{d} t \mathrm{d} x \\ &= \int_0^{\pi} \left(\int_t^{\pi} \mathrm{d} x \right) \frac{\sin t}{\pi - t} \mathrm{d} t \\ &= \int_0^{\pi} \sin t \mathrm{d} t \\ &= 2 \end{aligned}$$